

Practice Session 3 Answers

Part 1 : The Relationship Between Two Quantitative Variables/ Correlation and Regression

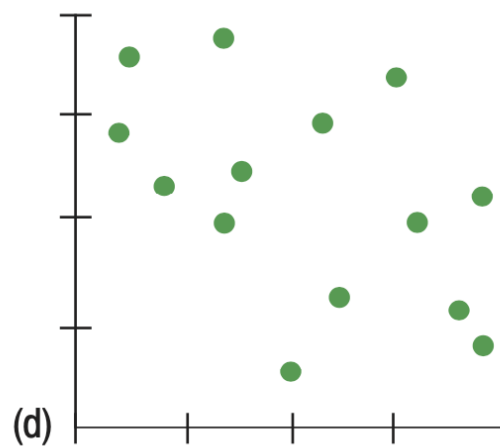
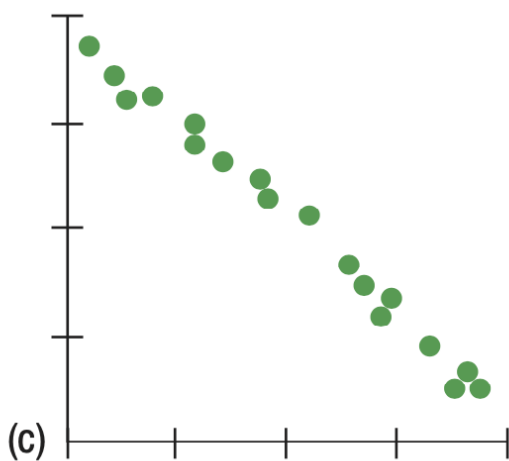
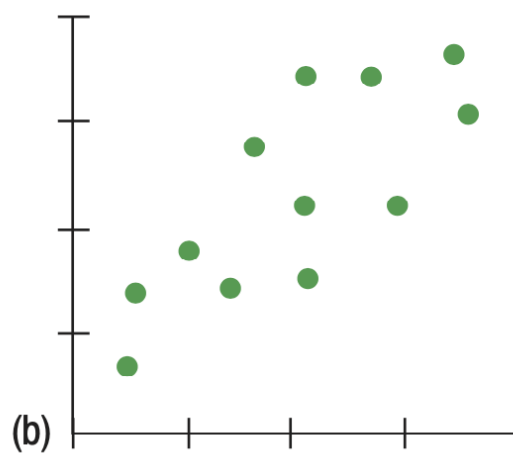
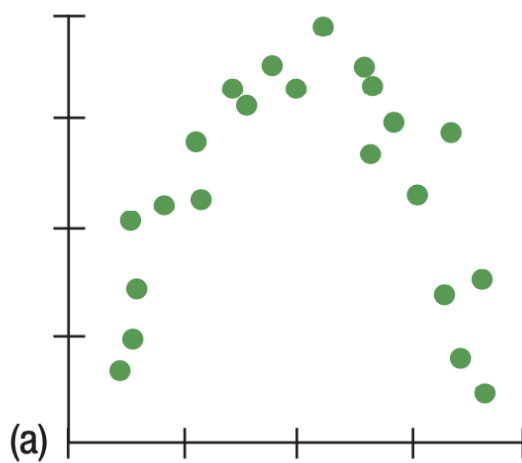
In this session you might use the formula of the correlation between two quantitative variables:

$$r_{xy} = \frac{1}{(n-1)s_x s_y} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

You may use the following R functions: `plot()`, `lm()`, `cor()`, `abline()`. And you might need to download Lock5Data using `library(Lock5Data)`.

1.1 Describe scatterplots and correlation

Here are several scatterplots. The calculated correlations are 0.006, - 0.977, - 0.487, and 0.777. Match each scatter plot with the appropriate correlation coefficient.



Answers:

a. 0.006

b. 0.777

c. -0.977

d. -0.487

1.2 Create scatterplots in R

Load the data `FloridaLakes` from `library(Lock5Data)`.

1. Describe the type of each of the variables `pH`, `Calcium`, and `Alkalinity`.
2. Create a three scatter plots for each pair of the variables: `pH` vs `Calcium`, `pH` vs `Alkalinity`, and `Calcium` vs `Alkalinity`. Add the main title to each plot.
3. What is the correlation coefficient between `pH` and `Calcium`. Is it positive or negative?
4. What do these coefficients mean in the context of this data ?
5. Try to calculate the correlation coefficient between `pH` and `Calcium` without using R function.

Answers:

```
# download the data and load it into R
library(Lock5Data)
data(FloridaLakes)
```

1. Describe the type of each of the variables `pH` and `Calcium`, `Alkalinity`.

The variables `pH` and `Calcium`, `Alkalinity` are quantitative variables (continuous)

2. Create a scatter plots for each of the pair of variables (add the main title of the plot).

```
# Create scatter pH vs Calcium
par(mfrow= c(1,3))

plot(FloridaLakes$pH,
     FloridaLakes$Calcium,
     main = "ph vs Calcium Scatter plot")

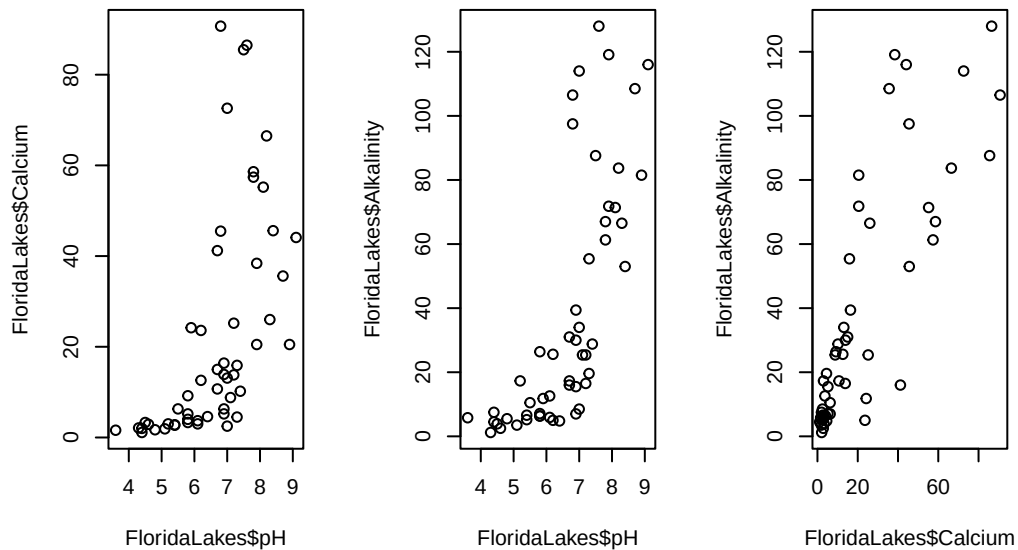
# Create scatter plot pH vs Alkalinity

plot(FloridaLakes$pH,
     FloridaLakes$Alkalinity ,
     main = "ph vs Alkalinity Scatter plot")

# Create scatter plot Calcium vs Alkalinity

plot(FloridaLakes$Calcium, FloridaLakes$Alkalinity,
     main = "Calcium vs Alkalinity Scatter plot")
```

ph vs Calcium Scatter plot ph vs Alkalinity Scatter plot Calcium vs Alkalinity Scatter



3. What is the correlation coefficient between pH and Calcium. Is it positive or negative.

```
# correlation

cor_pH_Cal <- cor(FloridaLakes$pH, FloridaLakes$Calcium)

cor_pH_Alkal <- cor(FloridaLakes$pH, FloridaLakes$Alkalinity)

cor_Cal_Alkal <- cor(FloridaLakes$Calcium, FloridaLakes$Alkalinity)

cor_pH_Cal
```

```
[1] 0.5771327
```

```
cor_pH_Alkal
```

```
[1] 0.7191657
```

```
cor_Cal_Alkal
```

```
[1] 0.8326042
```

4. What do these coefficients mean in the context of this data.

A correlation coefficient of 0.577 between pH and Calcium means that there is a linear positive association of moderate strength of 0.577.

5. Calculate the correlation coefficient between pH and Calcium without using R function.

```
## Method 2 to calculate the correlation using the formula

n <- length(FloridaLakes$pH)

X <- FloridaLakes$pH

Y <- FloridaLakes$Calcium

cor_pH_Cal <- sum((X - mean(X))/sd(X) * (Y - mean(Y))/sd(Y))/(length(X) - 1)

cor_pH_Cal
```

```
[1] 0.5771327
```

Part 2: The relationship between two Quantitative Variables / Linear Regression

Remember that the fitted regression line is defined by the equation:

- $\hat{y} = a + bx$, or
- $Response = a + b(Explanatory)$
- Residuals = observed - predicted

Where:

- y : the response (dependent) variable, the one we predict
- x : the explanatory (independent) variable
- a : is the y-intercept
- b : is the slope of the regression line

You may use the following R functions: `plot()`, `lm()`, `cor()`, `abline()`, `coef()`, `predict()`, `fivenum()`. If needed, install the `Lock5Data` package once with `install.packages("Lock5Data")`, then load it with `library(Lock5Data)`.

2.1 Exercise 1: State if the following sentences are true or false.

- a. We choose the linear model that passes through the most data points on scatter plot.
- b. The residuals are observed y-values minus the y-value predicted by linear model.
- c. Least square means that the square of the largest residuals is as small as it could possibly be.
- d. Some of the residuals from the least squares line will be positive and some will be negative.
- e. Least squares means that some of the squares of the residuals are minimized.
- f. We write $\hat{y}$ to denote the predicted values and y to denote the observed values.

Answers:

- a. False; we choose a model that minimizes the sum of squared residuals (i.e., sum of squared distances between each point and the regression line).
- b. True; this is the definition of a residual.
- c. False; the least squares method minimizes the **total** sum of squared distances, not the largest squared distance. It takes into account all observations, not a single observation.
- d. True; for a least-squares line that includes an intercept, the residuals always sum to zero. So unless every residual equals 0, some residuals must be positive and some must be negative.
- e. False; the least squares method minimizes the **total** sum of squared distances. The least squares method does not minimize individual squared residuals; it minimizes their sum.
- f. True; this is the definition of y and $\hat{y}$ in regression model.

2.2 Exercise 2:

Use data `FloridaLakes` and the two variables `Alkalinity` and `Calcium`.

```
# Load the Lock5Data package and the data
library(Lock5Data)
data(FloridaLakes)
```

1. Find the correlation for the `Alkalinity` and `Calcium`.
2. Using the appropriate R function, create a linear model to predict `Alkalinity` from `Calcium`.
3. Find the coefficients of the regression and write the correct equation of this linear model.
4. Interpret the intercept and slope of the model within the context of the data.
5. Predict the `Alkalinity` level when `Calcium` = 2.5.
6. The actual `Alkalinity` level was 8.5 when `Calcium` = 2.5. Find the residual for this data point.
7. Find the five number summary for the variable `Calcium`.
8. Predict the `Alkalinity` level when `Calcium` = 0.5.
9. Do both of these predictions make sense given the data?

Answers:

1. Find the correlation for the `Alkalinity` and `Calcium`.

```
alk <- FloridaLakes$Alkalinity
cal <- FloridaLakes$Calcium

cor_alk_cal <- cor(alk, cal)
cor_alk_cal
```

```
[1] 0.8326042
```

2. Using the appropriate R function, create a linear model to predict `Alkalinity` from `Calcium`.

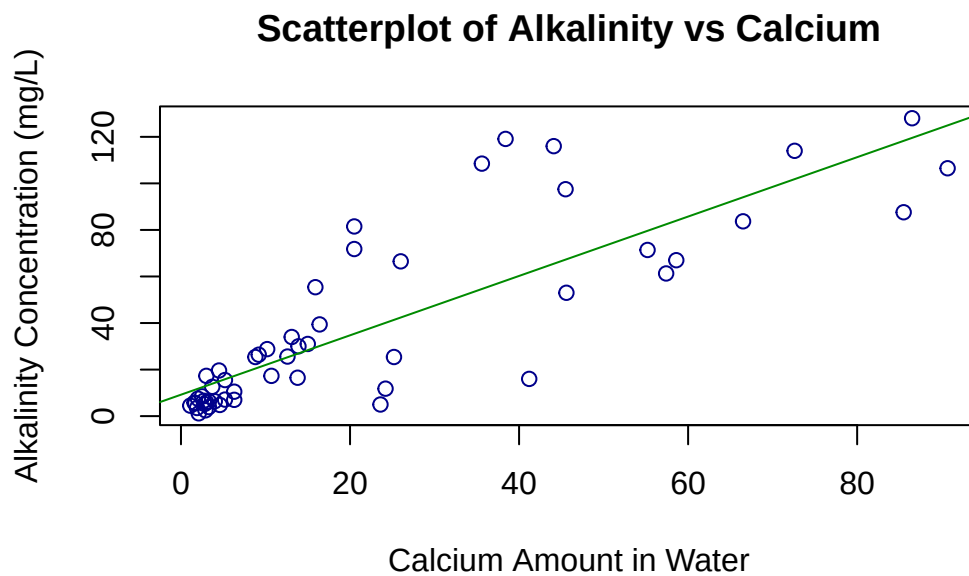
```
# Fit a linear regression model
alk_cal_fit <- lm(alk ~ cal) #data = FloridaLakes)
alk_cal_fit
```

```
Call:
lm(formula = alk ~ cal)
```

```
Coefficients:
(Intercept)      cal
      9.206      1.276
```

```
# Create a scatterplot of Alkalinity vs Calcium
plot(alk~ cal,          #data = FloridaLakes,
     main = "Scatterplot of Alkalinity vs Calcium",
     xlab = "Calcium Amount in Water",
     ylab = "Alkalinity Concentration (mg/L)",
     col = "blue4")

# Add the regression line to the plot
abline(alk_cal_fit, col = "green4")
```



3. Find the coefficients of the regression and write the correct equation of this linear model.

After running the following code you will get the intercept and the slope values of the regression line equation as follows:

$$\hat{Alkalinity} = 9.206 + 1.276 \cdot Calcium.$$


```
# you can use one of the following methods:
```

```
## method 1  
print(alk_cal_fit)
```

Call:

```
lm(formula = alk ~ cal)
```

Coefficients:

(Intercept)	cal
9.206	1.276

```
## or method 2  
reg_coef <- coef(alk_cal_fit)  
reg_coef
```

(Intercept)	cal
9.205522	1.275777

4. Interpret the intercept and slope of the model within the context of the data.

Slope: for each additional 1 mg/L of calcium, the predicted alkalinity increases by 1.276 mg/L, on average.

Intercept: when calcium is 0 mg/L, the predicted alkalinity is 9.206 mg/L. Note that Calcium = 0 is below the smallest observed value in the data (1.1 mg/L), so this intercept value is itself an extrapolation and should be interpreted with caution.

5. Predict the Alkalinity level when Calcium = 2.5.

```
# predict Alkalinity when Calcium = 2.5  
  
## method 1  
(predicted_alk_2.5 <- reg_coef[1] + reg_coef[2] * 2.5)
```

(Intercept)
12.39497

```
## method 2
(predicted_alk_2.5 <- 9.206 + 1.276 * 2.5)
```

```
[1] 12.396
```

```
# method 3: using predict() with newdata
(predicted_alk_2.5 <- predict(alk_cal_fit, newdata = data.frame(cal = 2.5)))
```

```
1
12.39497
```

6. The actual Alkalinity level was 8.5 when Calcium = 2.5. Find the residual for this data point.

```
#residual when Calcium = 2.5

# method 1: using the predicted value from Question 5
(res_2.5 <- 8.5 - predicted_alk_2.5)
```

```
1
-3.894966
```

7. Find the five number summary for the variable Calcium.

```
fivenum(cal)
```

```
[1] 1.1 3.3 12.6 35.6 90.7
```

8. Predict the Alkalinity level when Calcium = 0.5

```
(predicted_alk_0.5 <- reg_coef[1] + reg_coef[2] * 0.5)
```

```
(Intercept)
9.843411
```

9. Do both of these predictions make sense given the data?

The level of Calcium = 2.5 is within the range of the data, but the level of Calcium = 0.5 is an extrapolation.

Part 3: Review

3.1 Exercise 3:**

From the data `ICUAdmissions` create descriptive statistics (such as: `boxplot`, `histogram`, `five number summary`) for the variables `Infection`, `Age`, and `HeartRate`.

1. Create descriptive statistics using one quantitative variable.
2. Create descriptive statistics using one categorical variable.
3. Descriptive statistics for two quantitative variables.
4. Descriptive statistics for one quantitative and one categorical variable.

Answers:

```
# Load the Lock5Data package and the data
library(Lock5Data)
data(ICUAdmissions)
```

It seems that the variable types are:

- Age: quantitative variable
- Infection: categorical variable
- HeartRate: quantitative variable

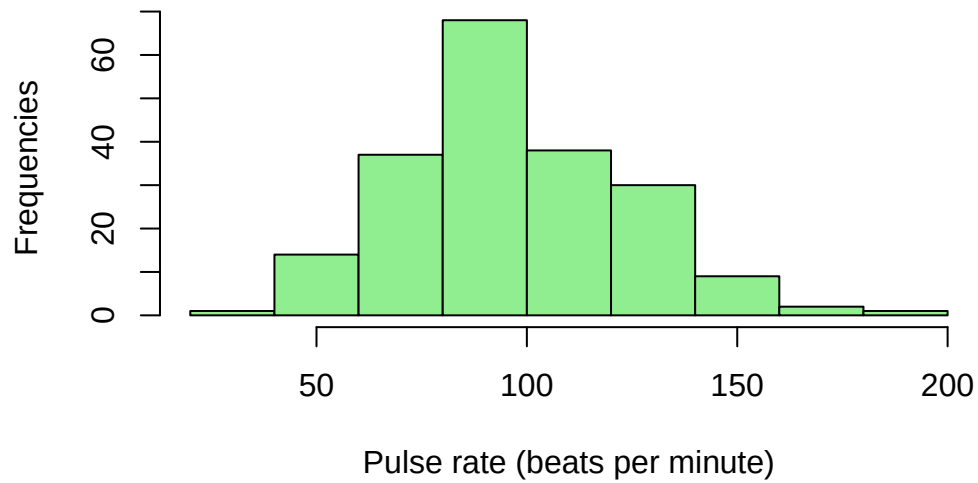
3.1 Descriptive stats for “one quantitative variable”

we can describe a quantitative variable with: `histogram`, `five number summary`, or `boxplot`

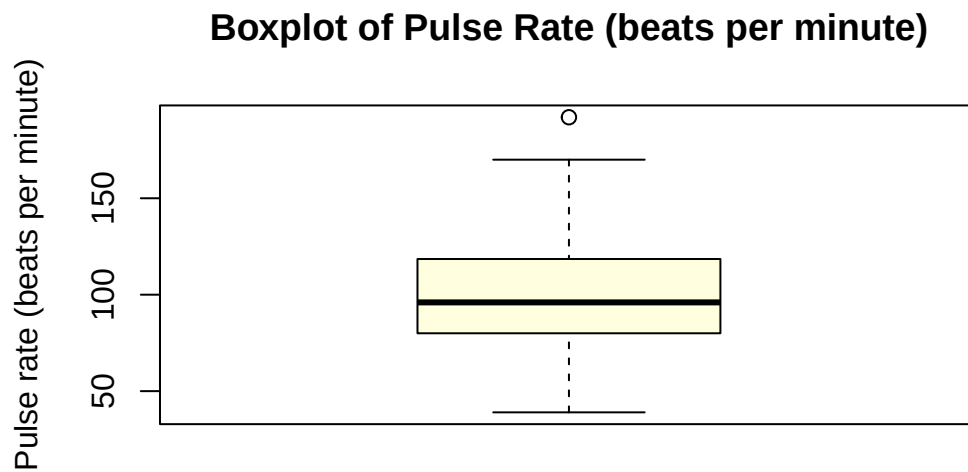
```
# Histogram for Heart Rate
heart_rate <- ICUAdmissions$HeartRate # Heart Rate - Pulse rate (beats per minute)

hist(heart_rate,
     main = "Histogram of Pulse Rate (beats per minute)",
     xlab = "Pulse rate (beats per minute)",
     ylab = "Frequencies",
     col = "lightgreen")
```

Histogram of Pulse Rate (beats per minute)



```
# Boxplot for Heart Rate
boxplot(heart_rate,
  main = "Boxplot of Pulse Rate (beats per minute)",
  ylab = "Pulse rate (beats per minute)",
  col = "lightyellow")
```



```
# Five-number summary for Heart Rate
five_numbers_heart_rate <- fivenum(heart_rate)
five_numbers_heart_rate
```

```
[1] 39.0 80.0 96.0 118.5 192.0
```

```
# As a second example of a quantitative variable, the five-number summary for Age
age <- ICUAdmissions$Age
five_numbers_age <- fivenum(age)
five_numbers_age
```

```
[1] 16.0 46.5 63.0 72.0 92.0
```

3.2. Descriptive stats for “one categorical variable”:

we can describe a categorical variable with: bar plot, table

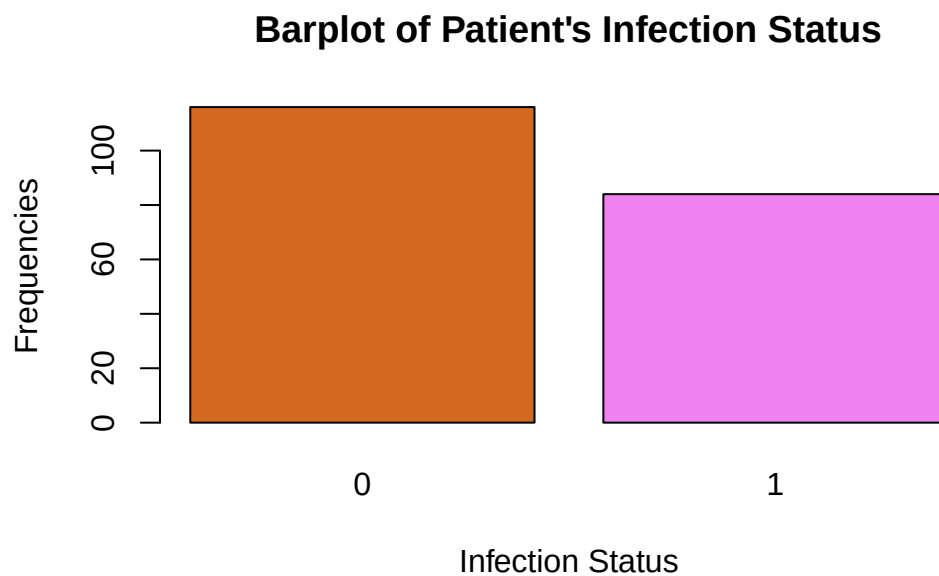
```
# one categorical variable such as (Infection)

## one-way table (Infection)
#infection <- factor(ICUAdmissions$Infection, labels = c("No infection", "infection"))

infection <- ICUAdmissions$Infection
infection_table <- table(infection)
infection_table
```

```
infection
  0    1
116  84
```

```
# create barplot
barplot(infection_table,
  main = "Barplot of Patient's Infection Status",
  xlab = "Infection Status",
  ylab = "Frequencies",
  col = c("chocolate", "violet"))
```



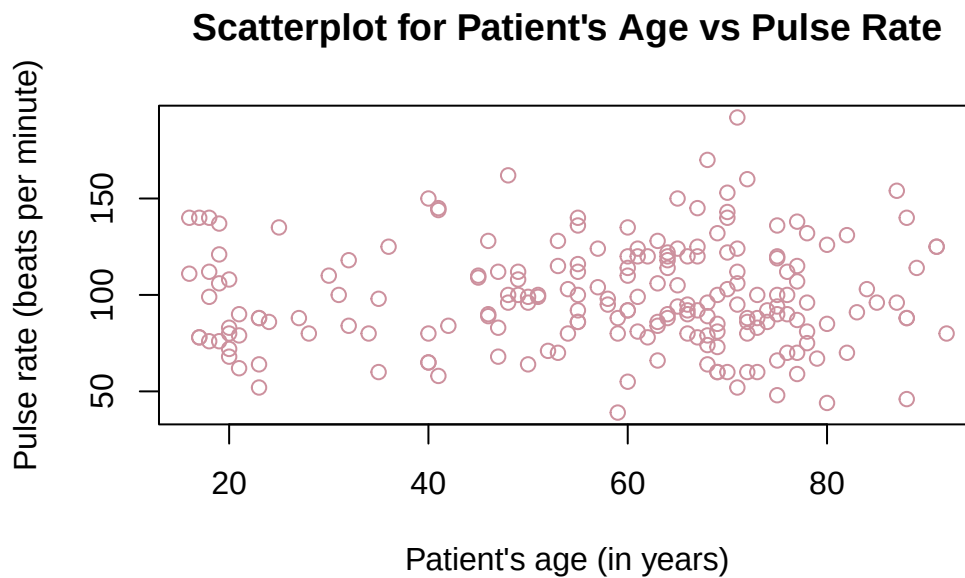
```
## one-way table proportions
(infection_props <- prop.table(infection_table))
```

```
infection
  0    1
0.58 0.42
```

3.3 Descriptive stats for “two quantitative variables”:

we can describe two quantitative variables with: scatter plot, correlation

```
# scatter plot for Age and HeartRate
plot(age, heart_rate,
     main = "Scatterplot for Patient's Age vs Pulse Rate",
     xlab = "Patient's age (in years)",
     ylab = "Pulse rate (beats per minute)",
     col = "pink3")
```



```
# correlation for age and heart rate  
cor(age, heart_rate)
```

```
[1] 0.03736843
```

3.4 Descriptive stats for “one quantitative variable vs one categorical variable”:

we can describe age (quantitative) versus Infection (categorical variable) with boxplot.

```
# boxplot of age (quantitative) versus categorical variable Infection (which has two levels)  
boxplot(age ~ infection,  
        main = "Boxplot of Patient's Age vs Infection Levels",  
        xlab = "Infection Levels",  
        ylab = "Patient's age (in years)",  
        col = c("orange1", "yellow3"))
```

